

VG4142SxxxX0M1 Wireless Module Hardware Specifications

V1.0



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1. Overview

VG4142SxxxX0M1 series wireless module, based on PANCHIP's PAN3031 high-performance wireless transceiver chip design, is an ultra-long-distance bidirectional wireless transceiver module with built-in power amplifier (PA) and low noise amplifier (LNA). PAN3031 is a low-power long-distance wireless transceiver chip using Chirp-IOT modulation and demodulation technology, supporting half-duplex wireless communication, with an operating frequency band of 400~510MHz/768~1020MHz. The chip has the characteristics of high anti-interference, high sensitivity, low power consumption and ultra-long distance. This module adds a power amplifier (PA) and a low noise amplifier (LNA) on the original basis, so that the maximum transmission power reaches 1W while the receiving sensitivity is further improved, and the overall communication distance and stability are greatly improved compared with ordinary power modules.

This series of modules integrates all RF-related functions and devices. Users do not need to have a deep understanding of RF circuit design, and can use the module to easily develop wireless solutions and wireless IoT devices with stable performance and high reliability.

Main features of the product:

- Chirp-IOT modulation
- Maximum link budget up to 164dB
- Maximum transmit power 30dBm, programmable configuration
- High receiving sensitivity: -134 dBm
- Wide operating voltage range: 4.5~5.5V
- Support bandwidth 125KHz, 250KHz, 500KHz
- Support spreading factor SF: 7~9

Applications:

- Supply chain and logistics
- Smart city
- Smart agriculture
- Industrial remote control
- Street light
- Remote control application
- Building automation
- Security systems

2. Technical Parameters

Parameter	Value range	Remark
Voltage range	4.5~ 5.5V	5V nominal
Frequency range	433MHz、868MHz、915MHz	The applicable frequency band is determined by the module model
Crystal frequency	32MHz	Passive crystal oscillator
Maximum output power	30dBm	1000mW
Air data rate	1.04kbps~20.4kbps	Programmable configuration
Modulation	Chirp- IOT	
Receiving sensitivity	- 134dBm	SF=9,BW=125kHz
Channel bandwidth	125kHz, 250kHz, 500kHz	Programmable configuration
TX current	600mA	Transmit power = 30dBm
RX current	30mA	Non-DC-DC Mode
Sleep current	<5uA	
Control interface	SPI	
Antenna impedance	50 Ohm	
Antenna connection	Side stamp hole	
Storage temperature	-55°C~+125°C	
Operating temperature	-40°C~+85°C	Industrial Grade
Dimensions	29.9x20.5mm	

3. Pin location diagram

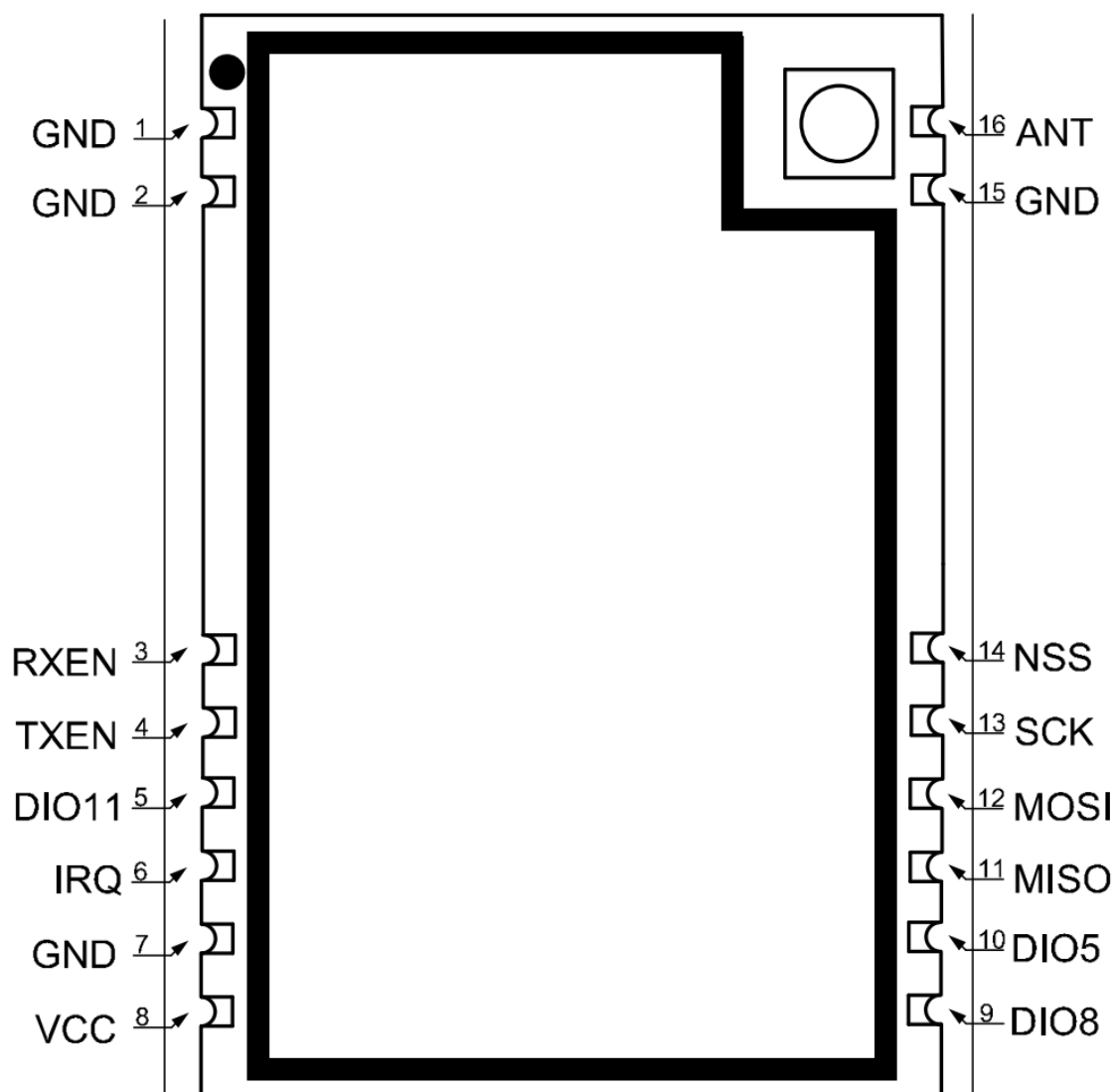


Figure 1-1 Top view

4. Pin Description

序号	引脚	类型	描述
1	GND	电源	地
2	GND	电源	地
3	RXEN	I	模块 LNA 放大器工作控制脚，接收时电平控制逻辑为 RXEN=1, TXEN=0; 休眠时 RXEN=0, TXEN=0
4	TXEN	I	模块 PA 放大器工作控制脚，发射时电平控制逻辑为 TXEN=1, RXEN=0; 休眠时 TXEN=0, RXEN=0
5	DI011	I/O	数字 IO，软件可配置，直连芯片 GPIO11
6	IRQ	O	中断信号脚
7	GND	电源	地
8	VCC	电源	电源正极
9	DI08	I/O	数字 IO，软件可配置，直连芯片 GPIO8
10	DI05	I/O	数字 IO，软件可配置，直连芯片 GPIO5
11	MISO	O	SPI 接口 MISO 数据输出
12	MOSI	I	SPI 接口 MOSI 数据输入
13	SCK	I	SPI 接口时钟输入
14	NSS	I	SPI 接口片选输入
15	GND	电源	地
16	ANT	I/O	RF 信号输入/输出，接 50 Ω 天线

5. Hardware Design Guidelines and Precautions

5.1. Hardware connection diagram

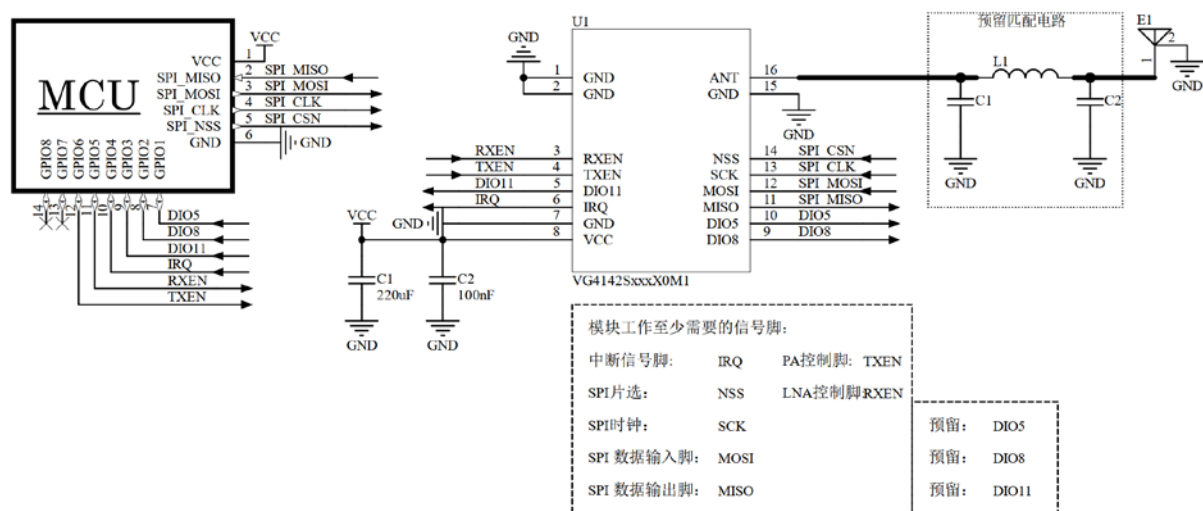


Figure 5-1 Programming development hardware connection

5.2 Power supply design and related considerations

- 1、请注意电源正负极的正确接法，并确保电源电压在推荐供电电压范围，如若超出模块最大允许供电范围，会造成模块永久损坏；模块电源脚的退耦电容尽量靠近模块电源引脚。
- 2、模块供电系统中，过大的纹波可能通过导线或者地平面耦合到容易受到干扰的线路上，例如天线、馈线、时钟线等敏感信号线上，容易引起模块的射频性能变差，所以我们推荐使用 LD0 或线性稳压器作为无线模块的供电电源。
- 3、选取 LD0 或线性稳压芯片时，需要注意电源的散热以及电源稳定输出电流的驱动能力；考虑整机的长期稳定工作，推荐预留 50%以上电流输出余量。
- 4、最好给模块单独使用一颗 LD0 或线性稳压器供电；如果采用 DC-DC 电源芯片，后面可以加一个 LD0 或线性稳压器作为模块电源的隔离，防止开关电源芯片的噪声干扰射频的工作性能。
- 5、MCU 与模块之间的通信线若使用 5V 电平，必须串联 1K-5.1K 电阻(不推荐，仍有损坏风险)。
- 6、射频模块尽量远离高压器件，因为高压器件的电磁波也会对射频信号产生一定的影响。
- 7、高频数字走线、高频模拟走线、大电流电源走线尽量避开模块下方，若不得已必须经过模块下方，需走线在摆放模块的 PCB 底板另一层，并保证模块下面铺铜良好接地。
- 8、本模块为大功率射频功率器件，最大功率发射时电流相对消耗较大，为确保其大功率发射驱动性能，需确保模块电源脚供电电压的稳定性及 PCB 底板电源走线的电流流通能力；另外模块发射工作时天线会产生较强的电磁场，

为减少对底板上 MCU 等敏感器件的电磁干扰，射频模块与天线尽量远离底板上的敏感器件。

5.3 Antenna Design Guidance

5.3.1 Stamp hole interface RF design

When the module RF output interface is selected as a stamp hole, use a 50ohm characteristic impedance trace to connect the antenna on the bottom PCB board during design. Considering the attenuation of high-frequency signals, it is necessary to pay attention to the bottom PCB RF trace length should be as short as possible. It is recommended that the longest trace length should not exceed 20mm, and the trace width should be continuous; when turning, try not to use sharp angles or right angles, and it is recommended to use arc lines.

首要推荐的射频走线转弯方式	
其次推荐的射频走线转弯方式	
比较糟糕的射频走线转弯方式，不推荐	

为尽量保证底板射频走线阻抗为 50 欧姆，可以根据不同板厚，按照如下参数进行调整。以下仿真值，仅供参考。

射频走线采用 20mil 线宽	板厚为 1.0mm 时，接地铺铜与走线间距为 5.3mil
	板厚为 1.2mm 时，接地铺铜与走线间距为 5.1mil
	板厚为 1.6mm 时，接地铺铜与走线间距为 5mil
射频走线采用 25mil 线宽	板厚为 1.0mm 时，接地铺铜与走线间距为 6.3mil
	板厚为 1.2mm 时，接地铺铜与走线间距为 6mil
	板厚为 1.6mm 时，接地铺铜与走线间距为 5.7mil
射频走线采用 30mil 线宽	板厚为 1.0mm 时，接地铺铜与走线间距为 7.6mil
	板厚为 1.2mm 时，接地铺铜与走线间距为 7.1mil
	板厚为 1.6mm 时，接地铺铜与走线间距为 6.6mil

5.3.2 Internal Antenna

Built-in antenna refers to the antenna that is welded on the PCB bottom plate and placed inside the product shell, including patch ceramic antenna, spring antenna, etc. When using built-in antenna, the product structure and the installation position of the antenna have a great influence on the RF performance. Under the premise that the product shell structure has enough space, the spring antenna should be placed vertically upward as much as possible; copper cannot be laid around the bottom plate where the antenna is placed, or the circuit board under the antenna can be hollowed out, because metal has a very strong absorption and shielding ability for RF signals, which will seriously affect the communication distance. In addition, the antenna should be placed on the edge of the bottom plate as much as possible.

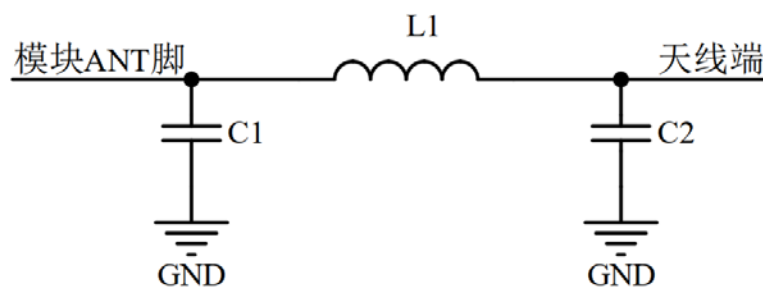
5.3.3 External Antenna

External antenna refers to the antenna installed outside the product housing through IPEX extension cable, SMA and other standard RF interfaces, including rod antenna, suction cup antenna, fiberglass antenna, etc. External antenna is basically a standard product. In order to better choose an antenna suitable for the module, the following should be noted when selecting antenna parameters:

1. The working frequency of the antenna should be consistent with the working frequency of the corresponding module.
2. The input characteristic impedance of the antenna should be 50ohm.
3. The antenna interface size should match that of the module.
4. The antenna standing wave ratio (VSWR) is recommended to be less than 2, and the antenna should have a suitable frequency bandwidth (covering the frequencies used in the actual application of the specific product).

5.3.4 Antenna Matching

The antenna is crucial to the transmission distance of the RF module. In practical applications, in order to facilitate the user's later antenna matching adjustment. It is recommended that the user reserve a simple type matching circuit between the antenna and the module ANT pin output when designing the schematic diagram. If the antenna is already a standard 50Ω, the component L1 is attached with a 0R resistor, and the components C1 and C2 do not need to be soldered. Otherwise, a network analyzer is needed to measure the actual impedance of the antenna and match it to determine the values of C1, L1, and C2. The routing from the module ANT pin to the antenna end should be as short as possible, and the longest routing length is recommended not to exceed 20mm.



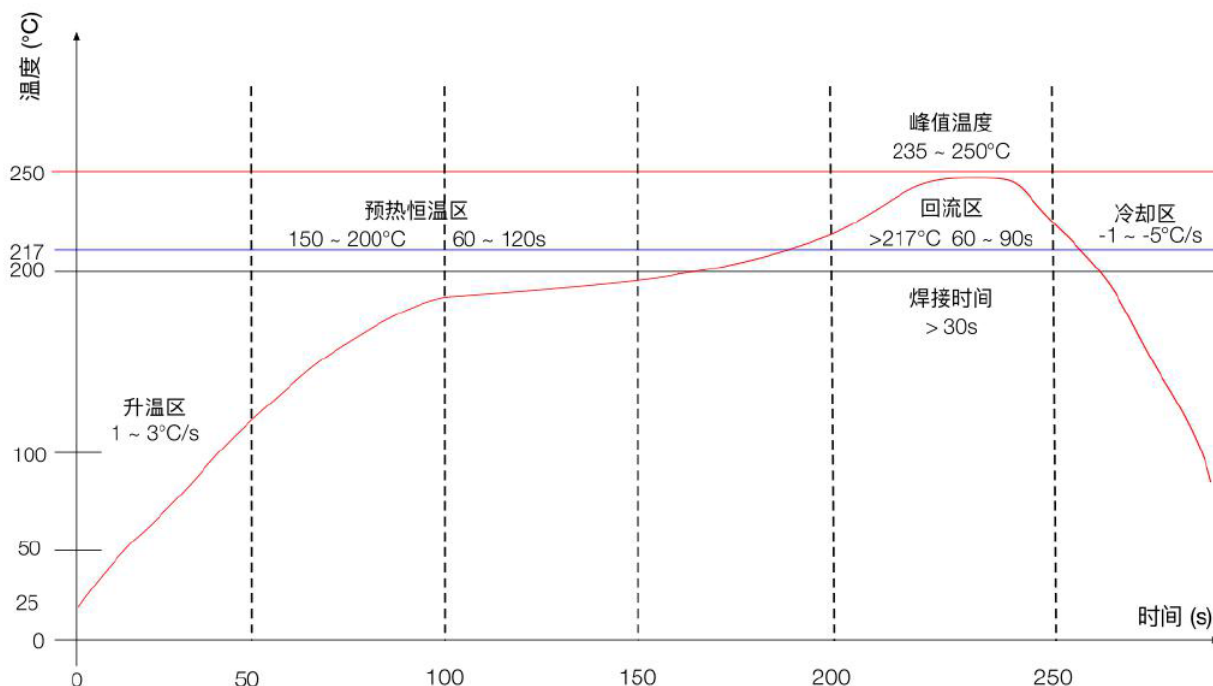
5-2 π -type matching circuit

6. Programming development notes

- 1) Since a PA amplifier is added to the chip, it is recommended that the output power of the front-stage PAN3031 chip be set to a maximum of 15dBm to achieve a PA saturation output power of 30dBm. After the PA reaches saturation power, its power output no longer increases with the increase of the front-stage power, but will increase the power consumption of the module.
- 2) In general, the receiving sensitivity of the RF chip is relatively poor at the integer multiple operating frequency of its crystal oscillator. It is recommended that users pay attention to avoiding the mirror frequency of the module crystal oscillator when selecting the operating frequency, that is, the integer multiple frequency of the crystal oscillator frequency. The crystal oscillator frequency of this module is 32MHz.
- 3) The TXEN and RXEN pins of the module are the logic control pins for controlling the PA&LNA devices inside the module. When using, please pay attention to the control level of the module TXEN and RXEN pins. The control logic is listed as follows:

Mode	TXEN	RXEN
Transmit	1	0
Receive	0	1
Sleep	0	0

7. Reflow Oven Curve



升温区 — 温度: 25 ~ 150°C 时间: 60 ~ 90s 升温斜率: 1 ~ 3°C/s
 预热恒温区 — 温度: 150 ~ 200°C 时间: 60 ~ 120s
 回流焊接区 — 温度: >217°C 时间: 60 ~ 90s; 峰值温度: 235 ~ 250°C 时间: 30 ~ 70s
 冷却区 — 温度: 峰值温度 ~ 180°C 降温斜率: -1 ~ -5°C/s
 焊料 — 锡银铜合金无铅焊料 (SAC305)

8. Warning of electrostatic damage

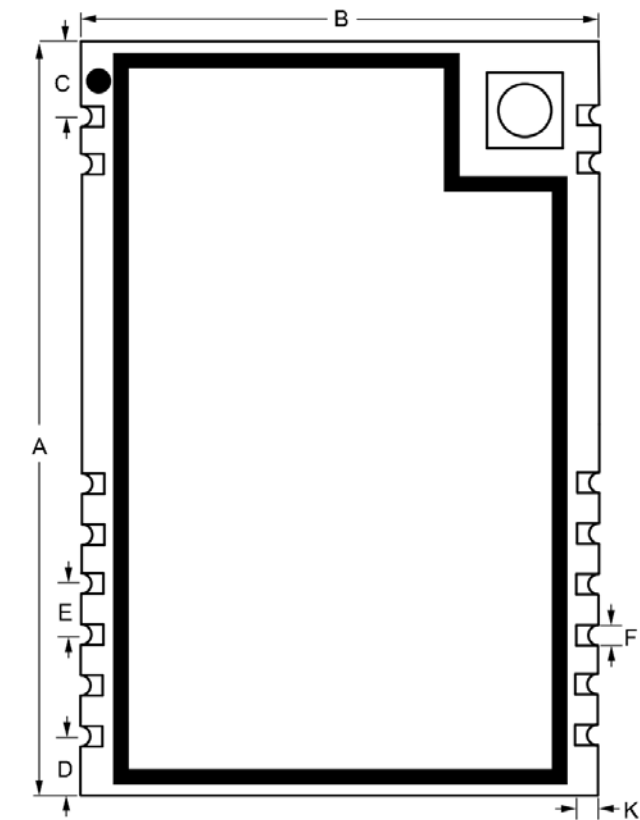
The RF module is a high-voltage electrostatic sensitive device. To prevent static electricity from damaging the module.

1. Strictly follow anti-static measures and do not touch the module with bare hands during the production process.
2. The module should be placed in an area that can prevent static electricity.
3. When designing the product, the anti-static protection circuit at the high-voltage input should be considered.

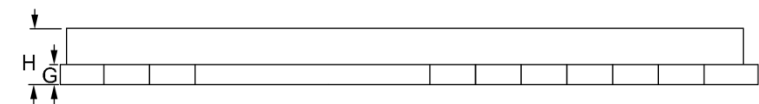


9. Packaging information

Mechanical dimensions (unit: mm)



编号	尺寸 (mm)	误差 (mm)
A	29.9	±0.5
B	20.5	±0.5
C	3.0	±0.1
D	2.3	±0.1
E	2.0	±0.1
F	1.2	±0.1
K	0.8	±0.1
G	1.0	±0.1
H	2.8	±0.2



10. Version update history

版本	更新内容	更新日期	维护人
V1.0	初始发布版本	2021 年 12 月 3 日	Dyming

11. Ordering information

序号	型号	说明
1	VG4142S433X0M1	433MHz 频段，编带包装\托盘包装
2	VG4142S868X0M1	868MHz 频段，编带包装\托盘包装
3	VG4142S915X0M1	915MHz 频段，编带包装\托盘包装

12. Statement

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